

Peak electrical power of a small vertical-axis wind turbine with 3D-printed textured blades

Wind-tunnel tests of one plain and three fuzzy-skin blade sets,
with rotor speed and surface roughness

Prepared by Stephen Eacuello and Taegu Kang
For Prof. Vahid Jahangiri, Prof. Yeonho Jeong, Prof. Manbir Sodhi
Tests 1 October 2026, URI wind tunnel
Date 5 October 2026
Data URI_VAWT_Texture_Data_2026-10-01.zip (accompanies this report)

Abstract

Fuzzy skin is a 3D-printing setting that roughens a part's outer surface by displacing its outer wall at random, up to a set thickness [1]. Roughness is known to change the performance of airfoils [2] and cylinders [3], but not of the thin curved-plate blades of the small vertical-axis rotor studied here. We fitted this rotor in turn with four printed blade sets: one without fuzzy skin (Plain) and three with fuzzy-skin thicknesses of 0.050 mm, 0.101 mm and 0.202 mm (FS 0.05, FS 0.10, FS 0.20). Each rotor was swept twice through 14 wind speeds from 10.2 to 38.0 m/s in the URI wind tunnel on 1 October 2026, with an electronic load measuring peak electrical power. Averaged over wind speed, FS 0.10 and FS 0.20 produced +12.8% [+10.1, +15.5] and +16.2% [+13.5, +19.0] more peak power than Plain (Tukey simultaneous 95% intervals). The FS 0.05 difference, +1.8% [−0.6, +4.2], was not resolved. The gains were largest at 21–25 m/s. At light load FS 0.10 and FS 0.20 also turned +6.6% and +9.1% faster than Plain, while the generator's voltage per rpm showed no resolved difference, so we infer that the gain arises in the rotor, not in the generator. Measured roughness (Ra, Pa) did not rank the sets by power; more surface measurement is needed to test whether it does. Each setting was one print, mounted once and run in order of increasing thickness, and Plain also had coarser layers (0.2 mm against 0.1 mm). The gains therefore cannot yet be attributed to fuzzy skin alone; replicate prints, remounting and an interleaved run order would settle this. Until then, rotors compared on this rig, including in the planned fault tests, should be printed with one recorded set of print settings.

1 Introduction

Printed blades carry surface texture even without fuzzy skin: the Plain set tested here has 0.20 mm layer lines. If texture changes rotor output, print settings must be controlled whenever rotors are compared, including in the fault tests planned for this rig (Section 8.1), and texture becomes a design variable. Fuzzy skin makes the texture adjustable without changing the CAD model: the slicer moves points of the outer wall in or out by a random distance up to the set thickness [1].

Roughness lowers the maximum lift and lift-to-drag ratio of airfoil sections [2] and moves the drop in a circular cylinder's drag to a lower Reynolds number [3]. The blades tested here are thin curved plates with square edges, at chord Reynolds numbers of 3.2×10^4 to 1.2×10^5 . Neither result predicts the effect on such blades, so it was measured. The tests address four questions:

1. Does peak electrical power differ between the blade sets? (Section 4.1, conclusion 1)
2. Do the differences depend on wind speed? (Section 4.2, conclusion 2)
3. Does a difference arise in the rotor or in the generator? (Section 4.3, conclusion 3)
4. Does measured surface roughness rank the sets by power? (Section 4.4, conclusion 4)

The report covers the 1 October 2026 session only. A fifth blade set run that day is excluded because its run log records its texture as unknown. The report also responds to two earlier requests: blade identity (Table 1) and an outline for a fault-testing plan (Section 8.1).

2 Test setup

2.1 Rotor and blade sets

The rotor has three straight blades attached at $R = 0.1016$ m from the axis, with span $H = 0.2451$ m (design values). Each blade is a thin curved plate, not an airfoil: it turns through 183° , with 48 mm chord, 1.79 mm wall and square edges. The swept area $A = 2RH = 0.0498 \text{ m}^2$ is nominal and is used only to normalise power. Each rotor is held by a mount we made: the rotor is inserted into the mount’s cylindrical shaft. The same mount, generator and bearings were used for every rotor.

Table 1 lists the four blade sets. A set mounted on the rig is called a rotor and named after the set. The College of Engineering’s I²(s) Print Lab printed all sets from one CAD model. Its Bambu Studio project specifies PLA on a Bambu Lab P1S with a 0.2 mm nozzle, 0.10 mm layers and four wall loops; fuzzy skin is painted on both faces of each blade at a 0.2 mm point distance, and only its thickness changes between sets. Plain has no slicer record; its measured layer period indicates 0.2 mm layers.

Table 1. Blade sets. Layer height from the measured layer period (Appendix B), except FS 0.20, whose texture hides the layer lines.

Rotor	Fuzzy skin	Layer height	Earlier label	Source of the settings
Plain	none	0.2 mm	smooth, baseline	Run log (“no fuzzy skin”); no slicer record.
FS 0.05	0.050 mm	0.1 mm	Ra 20	1 October run log; not in the slicer project.
FS 0.10	0.101 mm	0.1 mm	Ra 40	Slicer project, plate “40 RA”.
FS 0.20	0.202 mm	0.1 mm	Ra 80	Slicer project, plate “80 RA”.

2.2 Wind tunnel and measurements

The tunnel fan runs on an ABB ACS550 drive, commanded in rpm over Modbus (Fig. 1). Wind speed v is calculated from the fan speed N with the tunnel’s calibration (Appendix C):

$$v = 0.02132 N - 0.424 \quad (\text{m/s; } N \text{ in rpm}).$$

The fan-motor current at each set point agreed across all runs within 0.1 A, so the fan operating point did not change. Air temperature and pressure were not recorded, and no blockage correction is applied.

The generator’s three-phase output is rectified and drawn by a Chroma 63004-150-60 electronic load in constant-current mode. A Python host sets the current and records the terminal voltage V and current I at the end of each 1 s step; power is $P = VI$. All rotors were measured with the same instrument and settings.

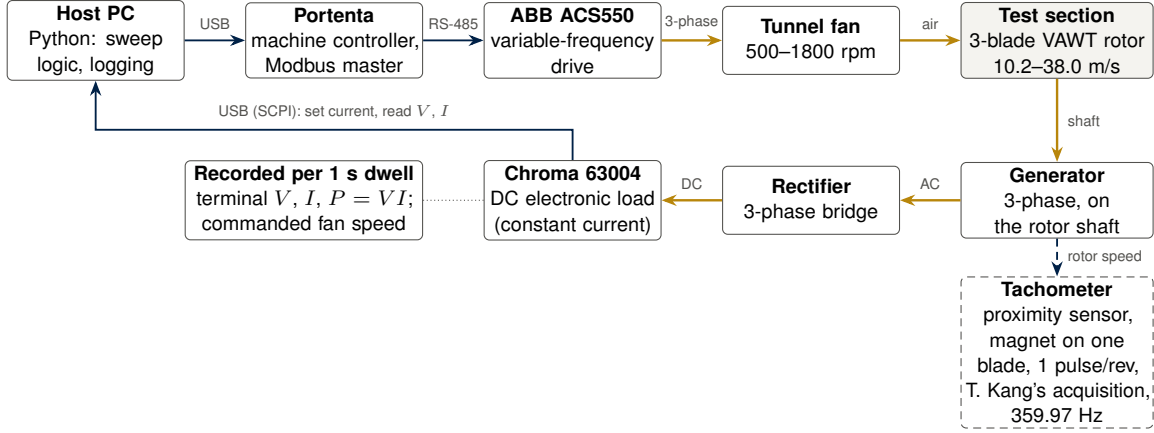


Figure 1. Measurement and control chain. The dashed path is T. Kang’s separate rotor-speed acquisition.

Rotor speed. The rig’s own speed input recorded nothing on 1 October. T. Kang recorded rotor speed with a proximity sensor triggered by a small magnet glued to one blade of each rotor, giving one pulse per revolution, sampled at 359.97 Hz. The magnet adds an unbalanced mass to that blade; its mass and position on each rotor were not recorded. His script (`RPM.m`, in the data package) reports for each set point the most frequent speed over single revolutions, n_0 , resolved to 0.7–4.0%. The load is at 0 A while the fan settles, so we take n_0 as the light-load speed. The rig’s data support this: the voltage at the first load step, V_1 , follows n_0 closely ($V_1/n_0 = 31.5 \text{ mV/rpm}$, coefficient of variation 1.7% over 104 values). At 500 rpm the load is already at 10 mA while the fan settles, so that set point is not used for speed. Speed during the load steps was not recorded in the summaries.

2.3 Test procedure

Each run visits 14 fan set points, 500–1800 rpm in 100 rpm steps. At each set point the host waits for the fan speed and the generator voltage to settle. The load then raises its current in small steps (20 mA at 1800 rpm, scaled with v^2 below), holding each for 1 s, until power has fallen below 80% of its maximum on two steps in a row. Each ladder of 13–30 steps traces power against current through its peak (Fig. 2). A run takes about 8 minutes.

Each rotor was mounted once and run twice without removal, in the order Plain, FS 0.05, FS 0.10, FS 0.20, from 15:50 to 18:24 (Table 2). 110 of the 112 ladders ended on the power roll-off; the other two ended after the peak, when the voltage fell below 0.5 V. One Plain blade broke at the end of run 2; its last ladder still matched run 1 within 0.1%. The rotor wobbled visibly on the mount at higher wind speeds. The wobble was not measured, and whether it differed between rotors was not recorded.

2.4 Surface measurement

Each set was scanned before the runs with a Keyence VR-6000 optical profilometer: one field of about $2.0 \times 1.4 \text{ mm}$ on one blade per set, two on FS 0.20 (Fig. 3). P_a is the mean deviation of the levelled height profile across the layer lines; R_a is the same after a Gaussian filter (ISO 16610-21 [5], cut-off 0.25 mm) removes the longer wavelengths. Appendix B gives the details.

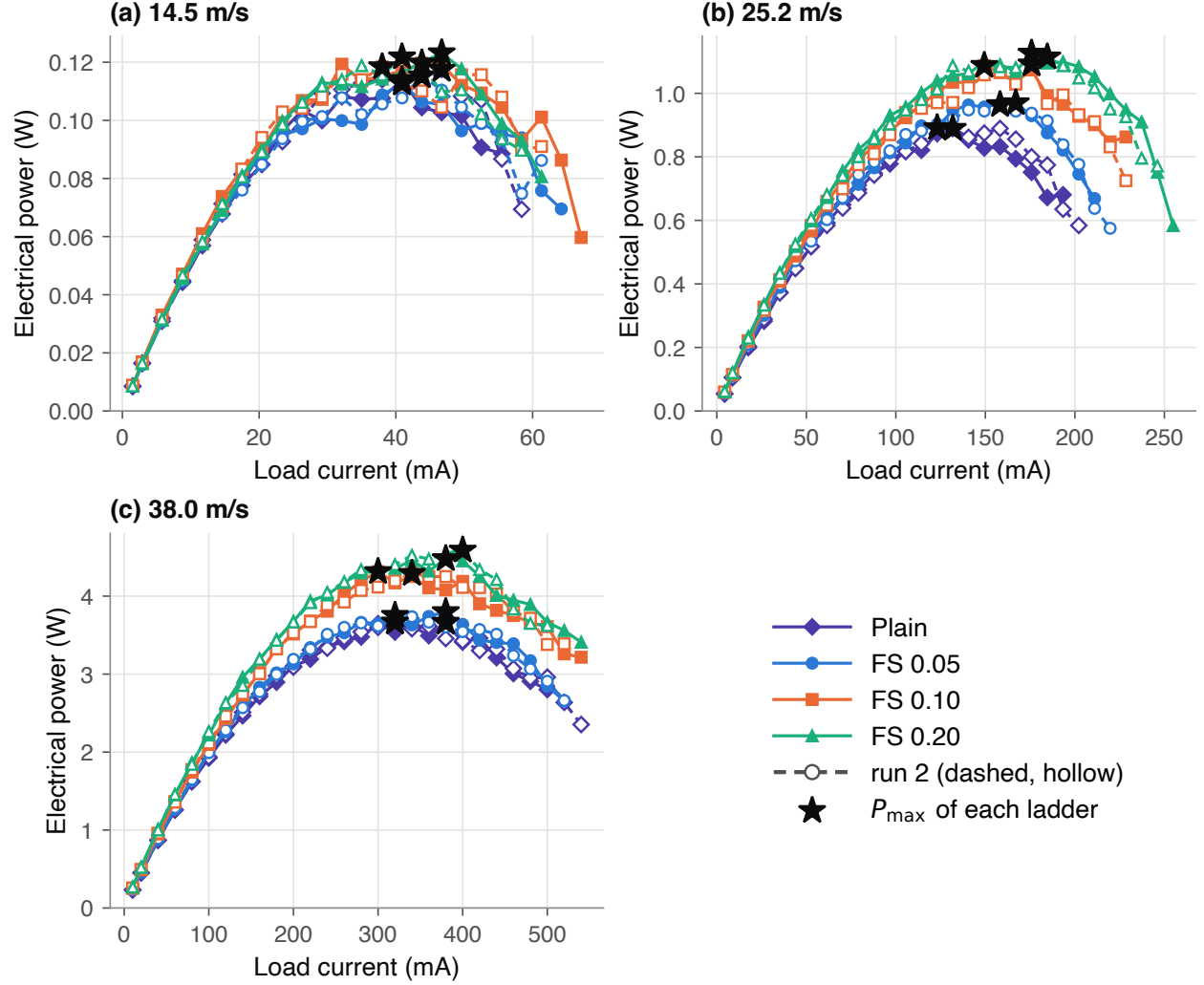


Figure 2. Measured load ladders at three wind speeds, all eight runs (solid: run 1; dashed, hollow: run 2). Stars mark $P_{\max}$, the largest step power.

3 Analysis

$P_{\max}$ is the largest step power in a ladder. Rotors are compared as ratios: for each run, the mean over the 14 set points of $\ln(P_{\max}/P_{\text{ref}})$, where P_{ref} is Plain's geometric mean at that set point. The eight run means are compared with one-way analysis of variance and Tukey's procedure for all six pairs [4]. A difference is *resolved* when its simultaneous 95% interval excludes zero. Whether the differences change with wind speed is tested by the rotor $\times$ set point interaction against the scatter between repeated runs. Details are in Appendix A.

To locate the source of a power difference, a straight line $V = V_{\text{oc}} - IR_{\text{int}}$ is fitted to each ladder. V_{oc} is the extrapolated open-circuit voltage; R_{int} is an apparent source resistance, which includes the winding, rectifier and wiring and also the rotor slowing as the load rises. Both, and the light-load speed n_0 , are compared between rotors as $P_{\max}$ is. The light-load tip-speed ratio is $\lambda_0 = 2\pi n_0 R / (60 v)$, and the electrical power coefficient is $C_{P,\text{el}} = P_{\max} / (\frac{1}{2} \rho A v^3)$ with standard air ($\rho = 1.204 \text{ kg/m}^3$); it includes generator and rectifier losses.

Table 2. Runs of 1 October 2026: local time of the first and last step, and the run mean of $P_{\max}$ relative to Plain.

Rotor	Run	Time	Change (%)	Data file stem
Plain	1	15:50–15:57	−0.4	v1_smooth_20261001
	2	16:50–16:58	+0.4	v1_smooth_repeat_20261001
FS 0.05	1	17:18–17:25	+2.3	v1_Ra20_20261001
	2	17:27–17:34	+1.3	v1_Ra20_repeat_20261001
FS 0.10	1	17:42–17:50	+12.4	v1_Ra40_20261001
	2	17:51–17:59	+13.1	v1_Ra40_repeat_20261001
FS 0.20	1	18:07–18:15	+15.7	v1_Ra80_20261001
	2	18:16–18:24	+16.8	v1_Ra80_repeat_20261001

4 Results

4.1 Peak power

Peak power rises from 0.030–0.034 W at 10.2 m/s to 3.67–4.53 W at 38.0 m/s, and $C_{P,el}$ from 0.09% to 0.28% (Fig. 4).

Averaged over wind speed, FS 0.10 and FS 0.20 produced +12.8% [+10.1, +15.5] and +16.2% [+13.5, +19.0] more peak power than Plain, and FS 0.05 +1.8% [−0.6, +4.2] (Fig. 5, Table 3). The rotor effect is far larger than the run-to-run scatter ($F_{3,4} = 324$, $p < 0.001$), and each rotor’s two runs agree within 1%. Five of the six pairs are resolved; FS 0.05 vs Plain is not. The smallest resolved difference, FS 0.20 over FS 0.10 (+3.0% [+0.6, +5.5]), is only just resolved. The intervals measure repeatability on one mounting. The FS 0.10 and FS 0.20 gains over Plain would stay resolved for a mounting-to-mounting variation up to 2.0% and 2.6%; FS 0.20 over FS 0.10 only up to 0.3%.

Table 3. Pairwise comparisons of $P_{\max}$, geometric mean over wind speeds: Tukey simultaneous 95% intervals and adjusted p -values. Last column: the same with a parabolic fit through the peak instead of the largest step.

Comparison	Change (%)	95% CI (%)	p	Parabolic fit (%)
FS 0.05 vs Plain	+1.8	[−0.6, +4.2]	$p = 0.12$	+1.6 [−1.2, +4.4]
FS 0.10 vs Plain	+12.8	[+10.1, +15.5]	$p < 0.001$	+13.0 [+10.0, +16.2]
FS 0.20 vs Plain	+16.2	[+13.5, +19.0]	$p < 0.001$	+16.2 [+13.1, +19.5]
FS 0.10 vs FS 0.05	+10.8	[+8.2, +13.5]	$p < 0.001$	+11.3 [+8.3, +14.4]
FS 0.20 vs FS 0.05	+14.2	[+11.5, +16.9]	$p < 0.001$	+14.4 [+11.3, +17.6]
FS 0.20 vs FS 0.10	+3.0	[+0.6, +5.5]	$p = 0.023$	+2.8 [+0.03, +5.7]

4.2 Dependence on wind speed

The gains change with wind speed by more than the scatter between repeated runs ($F_{39,52} = 9.2$, $p < 0.001$; Fig. 6). They are largest at 21–25 m/s (chord Reynolds number $6.6\text{--}8.0 \times 10^4$), where FS 0.20 peaks at +36.2% and FS 0.10 at +22.4%. The power of FS 0.10 and FS 0.20 also rises most steeply at a lower wind speed (22 and 20 m/s) than Plain’s (24–26 m/s).

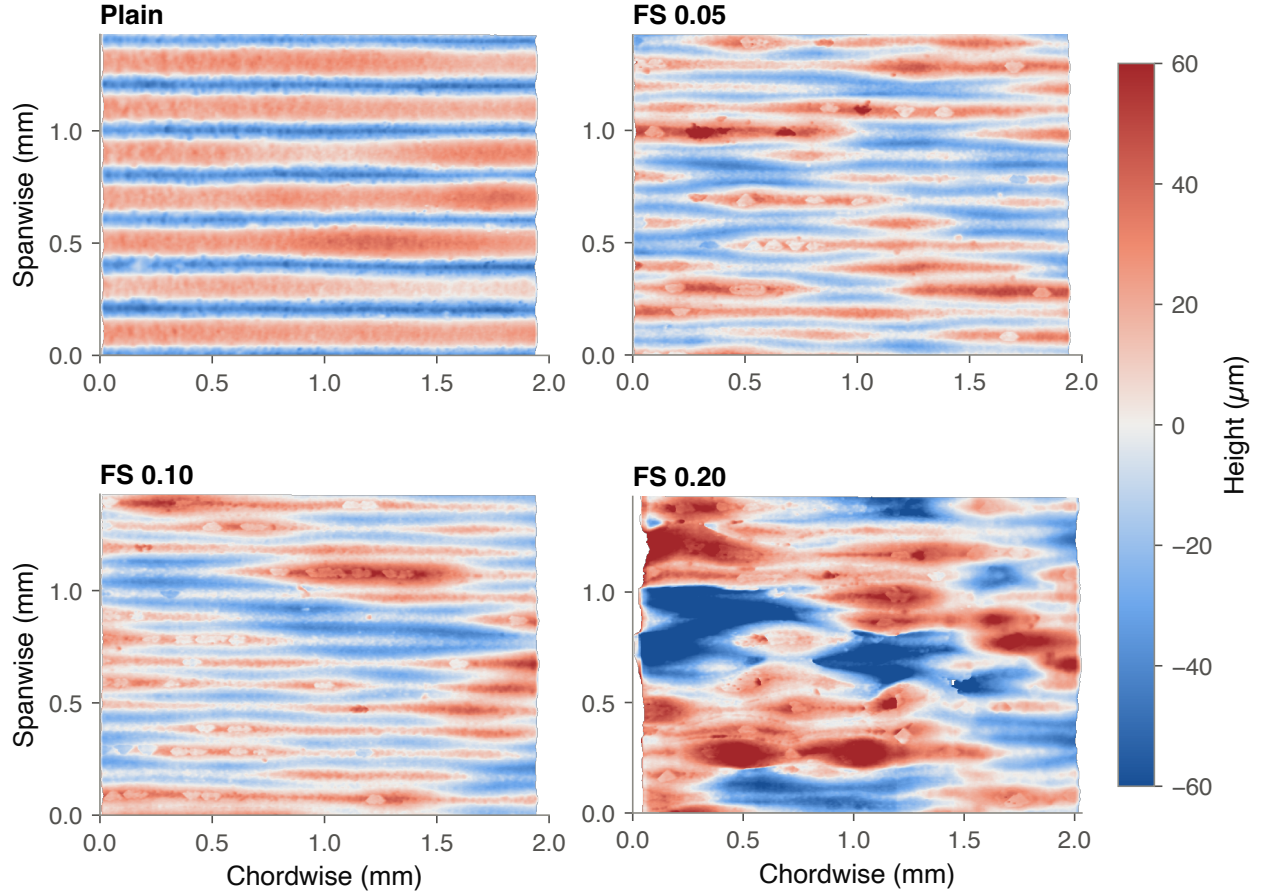


Figure 3. Profilometer height maps, one field per blade set. Layer lines run horizontally. Colour is clipped at $\pm 60 \mu\text{m}$, which FS 0.20 exceeds over 8% of its area.

4.3 Rotor or generator

Averaged over wind speed, V_{oc} of FS 0.10 and FS 0.20 differs from Plain's by +6.4% and +8.5%, while R_{int} shows no resolved difference (Table 4). Because the peak power of the fitted source scales with V_{oc}^2 , the higher V_{oc} alone accounts for the gains (+13.1% and +17.7% against +12.8% and +16.2% measured). Where the gains are largest, a lower R_{int} supplies up to 43% of the gain.

At light load, FS 0.10 and FS 0.20 turn +6.6% [+5.1, +8.1] and +9.1% [+7.6, +10.6] faster than Plain (Fig. 7). Each speed change matches the change in V_1 and V_{oc} within 0.7 percentage points, and the voltage per rpm shows no resolved difference between rotors (largest +0.7% [−0.7, +2.1]). The speed gain is larger from 20.9 m/s, where the band of largest power gains begins: 4–11% and 9–16%, against 2–7% and 1–6% below it.

4.4 Surface roughness

Neither Pa nor Ra ranks the rotors by power (Table 5). Plain has the highest Ra and the lowest power. FS 0.05 and FS 0.10 differ by only $0.7 \mu\text{m}$ in Pa and $1.2 \mu\text{m}$ in Ra, less than the two FS 0.20 fields differ from each other, yet their power differs by +10.8% [+8.2, +13.5]. Ra is also not like-for-like across the sets: the 0.25 mm cut-off keeps only about 65% of Plain's coarser layer lines (Appendix B).

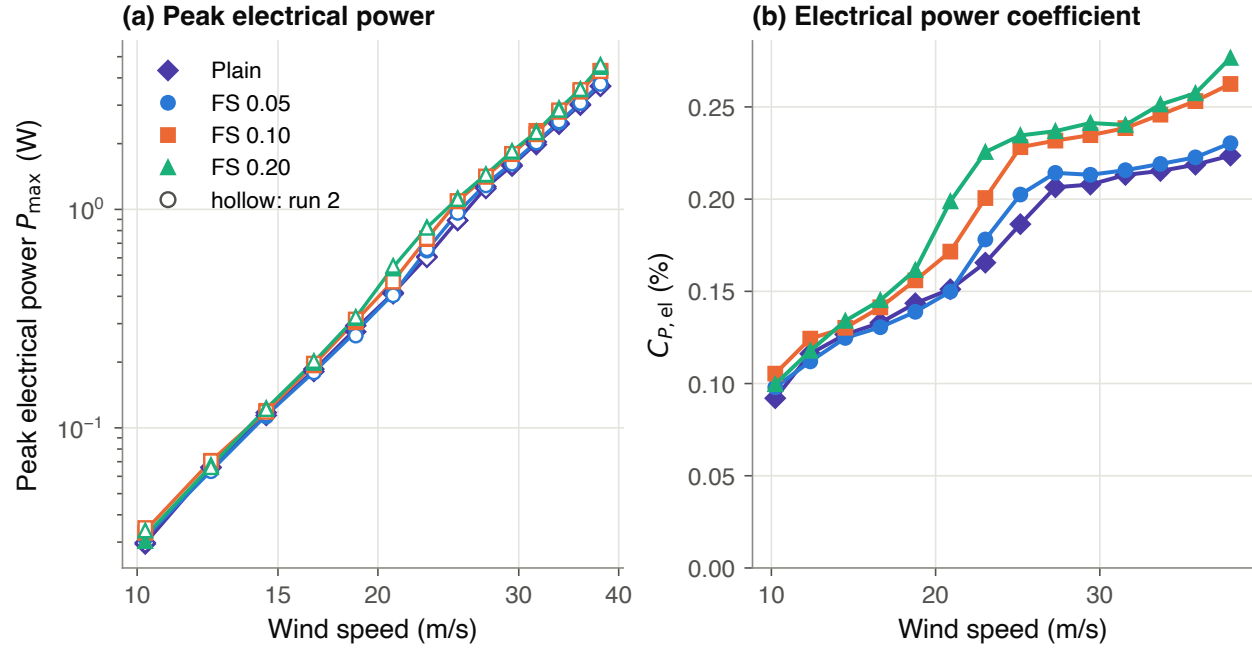


Figure 4. (a) Peak electrical power of every run (filled: run 1; hollow: run 2); lines join the geometric means of the two runs. (b) Electrical power coefficient, standard air.

Table 4. Thévenin parameters and light-load speed n_0 relative to Plain, geometric mean over wind speeds (n_0 from 600 rpm), with Tukey simultaneous 95% intervals; $C_{P,el}$ and λ_0 at 38.0 m/s.

Rotor	Change relative to Plain (%)			At 38.0 m/s	
	V_{oc}	R_{int}	n_0	$C_{P,el}$ (%)	λ_0
Plain	reference	reference	reference	0.22	0.21
FS 0.05	+1.2 [+0.2, +2.2]	-0.1 [-1.5, +1.4]	+1.5 [+0.1, +2.9]	0.23	0.21
FS 0.10	+6.4 [+5.3, +7.4]	-0.1 [-1.5, +1.4]	+6.6 [+5.1, +8.1]	0.26	0.22
FS 0.20	+8.5 [+7.4, +9.6]	+0.2 [-1.3, +1.6]	+9.1 [+7.6, +10.6]	0.28	0.24

Table 5. Surface roughness and peak power by rotor. FS 0.20: mean of two fields.

Rotor	Fuzzy skin (mm)	Pa (μm)	Ra (μm)	$P_{\max}$ change ^a (%)	$P_{\max}$ at 38.0 m/s (W)
Plain	none	19.0	12.7	reference	3.67
FS 0.05	0.050	13.9	9.4	+1.8 [-0.6, +4.2]	3.78
FS 0.10	0.101	13.2	8.2	+12.8 [+10.1, +15.5]	4.30
FS 0.20	0.202	28.7	10.3	+16.2 [+13.5, +19.0]	4.53

^aGeometric mean over wind speeds of the ratio to Plain, with the Tukey simultaneous 95% interval.

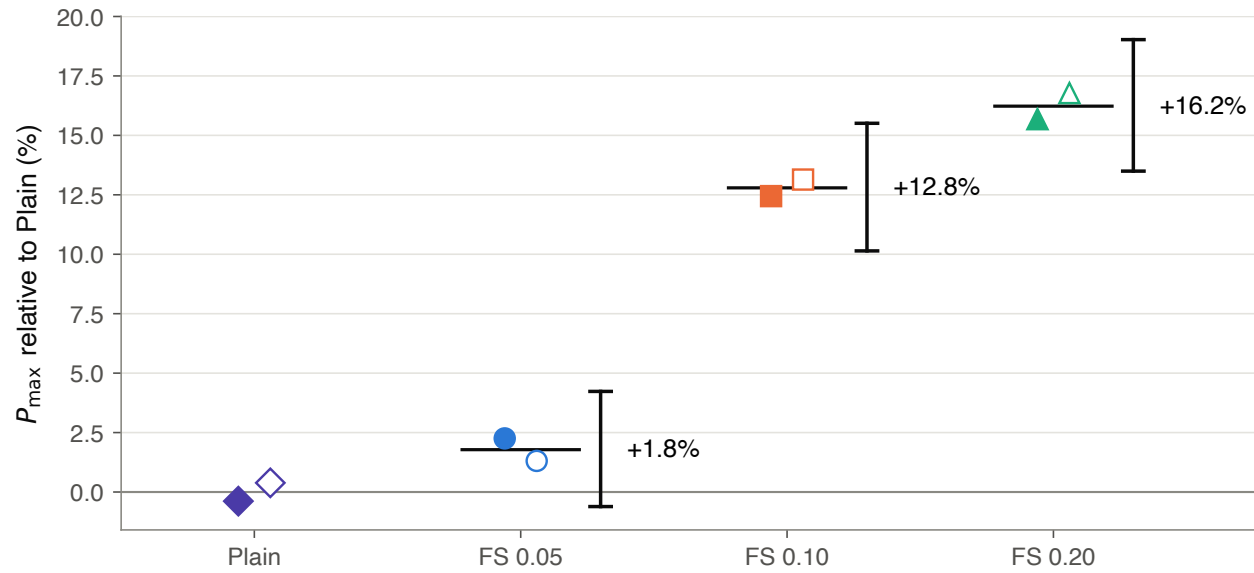


Figure 5. Each run's mean $P_{\max}$ relative to Plain (filled: run 1; hollow: run 2), with the Tukey simultaneous 95% interval for the difference from Plain.

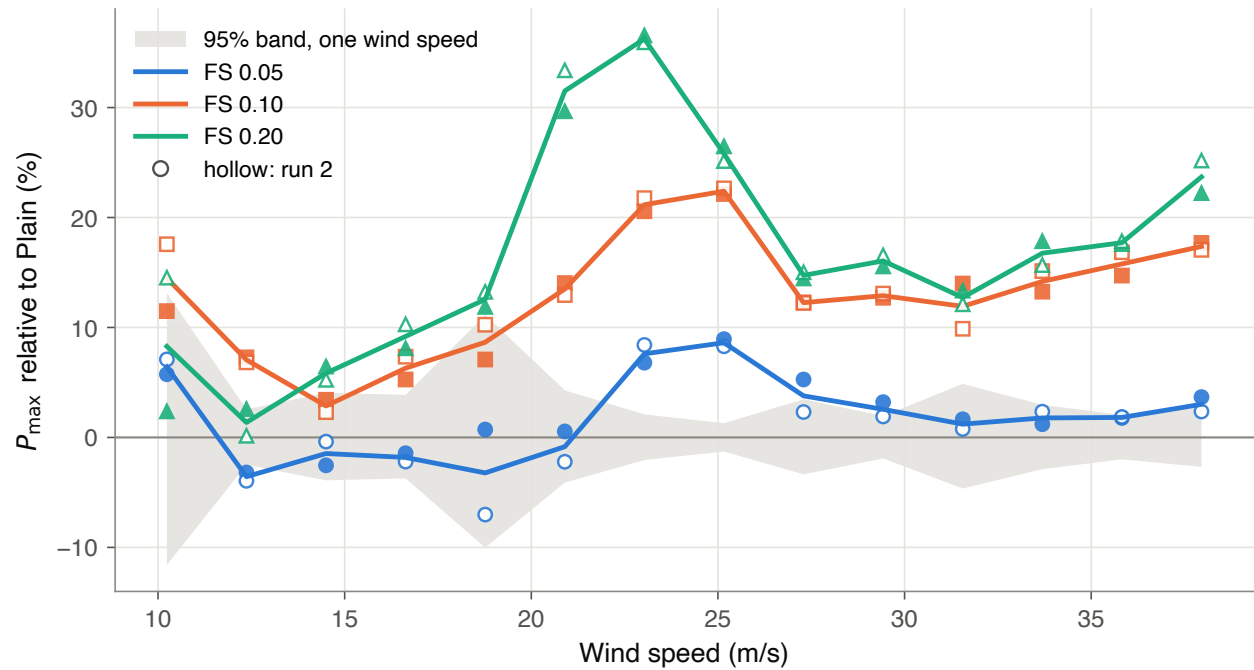


Figure 6. $P_{\max}$ of each textured rotor relative to Plain at each wind speed. Markers: individual runs (filled: run 1; hollow: run 2); lines: geometric mean; band: 95% interval of a single-wind-speed difference from run-to-run scatter.

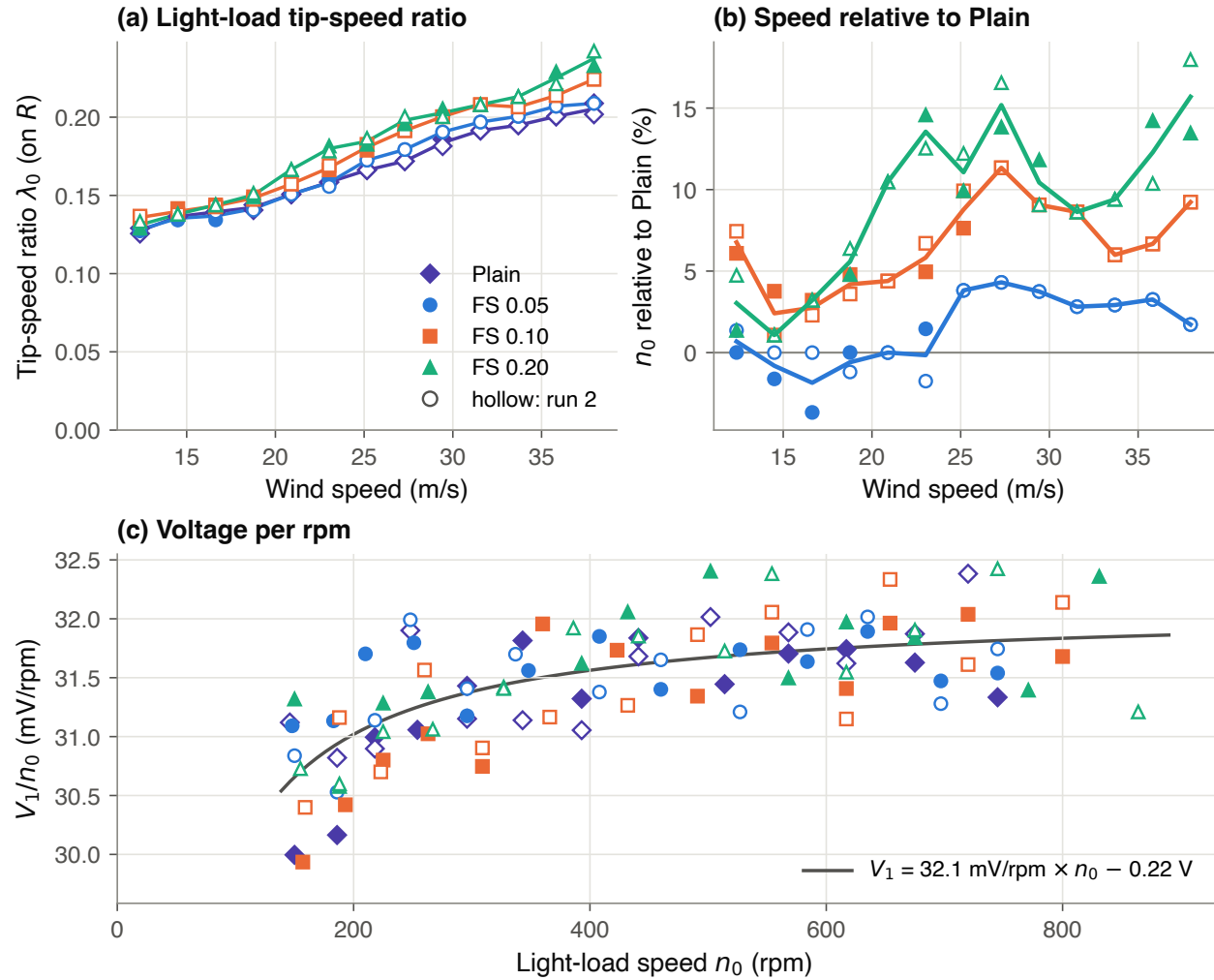


Figure 7. Light-load rotor speed n_0 from the tachometer, 600–1800 rpm (filled: run 1; hollow: run 2; lines: geometric mean). (a) Tip-speed ratio λ_0 at the attachment radius. (b) n_0 relative to Plain. (c) First-step voltage per rpm, V_1/n_0 , with the least-squares line $V_1 = a + bn_0$.

5 Discussion

Where the gain arises. We infer that the textured rotors produce a higher open-circuit voltage because they turn faster: the voltage per rpm is the same for all rotors to within 2.1%. With the same source resistance, they also turn faster at equal load current, that is, at about equal generator torque. The data do not show whether this comes from more aerodynamic torque or less resisting torque. All rotors used the same mount, generator and bearings, so less resisting torque would have to come from the blade sets themselves (mass, balance or mounting, none measured) or from a change in the shared parts during the session. Two mechanical effects are candidates: the tachometer magnet unbalances one blade, and the rotor wobbled at higher wind speeds. Either could change the resisting torque and differ between mountings.

Why the gains depend on wind speed. The gains peak at chord Reynolds numbers of $6.6\text{--}8.0 \times 10^4$, and the textured rotors' power rises most steeply at a lower wind speed than Plain's. A flow transition that texture moves to a lower Reynolds number could produce this pattern, as roughness does on a circular cylinder [3]. The flow was not examined, so this remains a hypothesis.

Roughness does not explain the ranking yet. R_a and P_a from one field per set may not represent the blades, may not capture the surface property that matters for this flow, or the power differences may not come from texture at all. More surface measurement is needed to test whether roughness ranks power (Recommendation 5).

6 Limitations

- **One print, one mounting.** Each texture level is one print mounted once, so print and mounting differences are confounded with texture. Plain also has coarser layers (0.2 mm); the three fuzzy-skin sets share 0.1 mm layers, so their comparisons (FS 0.10 and FS 0.20 over FS 0.05: +10.8% and +14.2%) are not affected by layer height.
- **Run order.** The rotors ran in order of increasing texture. Fitting a linear drift over the session (+0.8% per hour, [−1.7, +3.3]) leaves FS 0.10 and FS 0.20 resolved from Plain (+11.5% and +14.5%) but could explain the FS 0.05 difference. A step change between rotors cannot be separated from a rotor effect.
- **Mechanical.** The tachometer magnet on one blade unbalances each rotor, and the rotor wobbled on the mount at higher wind speeds. Neither was measured, so neither can be excluded as a cause of the differences.
- **Other.** Rotor speed is known only at light load. Wind speed comes from the tunnel calibration (Appendix C), which affects absolute wind speed and $C_{P,el}$ but not the comparison between rotors, made at identical fan speeds. Air density was not recorded, and no blockage correction is applied.

7 Conclusions

1. FS 0.10 and FS 0.20 produced +12.8% [+10.1, +15.5] and +16.2% [+13.5, +19.0] more peak electrical power than Plain, averaged over 10.2–38.0 m/s. The FS 0.05 difference, +1.8% [−0.6, +4.2], was not resolved.
2. The gains were largest at 21–25 m/s, where FS 0.20 reached +36.2%.

3. The gains came with a higher open-circuit voltage from faster rotation (+6.6% and +9.1% at light load), with no resolved change in the generator's voltage per rpm. We infer that they arise in the rotor, not in the generator.
4. Ra and Pa from one field per set did not rank the rotors by power; more surface measurement is needed to test whether roughness does.
5. With one print and one mounting per texture, a fixed run order, coarser layers on Plain and unmeasured mechanical effects (magnet imbalance, wobble), the gains cannot yet be attributed to fuzzy skin alone.

8 Recommendations

1. **Replicate and randomise.** Print at least two sets per thickness, Plain included, from one saved project with 0.1 mm layers. Weigh and label every blade. Mount each set twice, run in an interleaved order, and re-run Plain after every third rotor.
2. **Stiffen and balance the mount.** Extend the cylindrical shaft that the rotor is inserted into, to reduce the wobble at higher wind speeds. Balance the tachometer magnet with equal masses on the other two blades, or move it to the shaft. Connect the rig's tunnel node, which logs vibration and ambient air, on every run.
3. **Log rotor speed at every step,** on the rig's own pulse input or on T. Kang's acquisition synchronised to the rig clock. This gives the power coefficient against tip-speed ratio along each ladder. A spin-down test of each mounted rotor would separate aerodynamic from resisting torque.
4. **Extend the wind calibration** over the full test range, and record air temperature and pressure on every run.
5. **Measure the surfaces more fully:** several fields per set, at fixed locations on two blades and both faces, with evaluation lengths of at least five cut-offs. This is needed to test whether roughness ranks the rotors by power.

8.1 Toward the fault-testing plan

Prof. Jeong asked for a fault-testing plan agreed before the next round. The rig gives an automated sweep of about 8 minutes, with a run-to-run repeatability of 0.58% in the run mean and 1.9% at a single set point. Blade sets printed with different settings differed by up to +16.2%, so fault blades and their reference set should be printed from one saved project, with the reference re-run between fault cases. The reference rotor should be balanced first, because the tachometer magnet already unbalances one blade. Candidate faults are a tip mass on one blade (imbalance), a chipped, shortened or missing blade, added bearing drag, and an open or high-resistance generator phase. Whether V_{oc} and R_{int} can tell rotor faults from electrical ones is untested; the first cases should test this with rotor speed logged at every step. We propose a short meeting to agree the fault list, severity levels and acceptance criteria before any fault blades are printed.

Acknowledgement

T. Richards modelled the blade, A. Muszynski helped set up the fuzzy-skin print settings, and the College of Engineering's I²(s) Print Lab printed the blade sets.

References

- [1] Prusa Research, “Fuzzy skin,” PrusaSlicer knowledge base, https://help.prusa3d.com/article/fuzzy-skin_246186, accessed 5 October 2026.
- [2] B. J. Wilcox, E. B. White and D. C. Maniaci, *Roughness Sensitivity Comparisons of Wind Turbine Blade Sections*, Sandia National Laboratories, SAND2017-11288, 2017, doi:10.2172/1404826.
- [3] E. Achenbach, “Influence of surface roughness on the cross-flow around a circular cylinder,” *Journal of Fluid Mechanics*, 46(2), 321–335, 1971, doi:10.1017/S0022112071000569.
- [4] D. C. Montgomery, *Design and Analysis of Experiments*, 9th ed., Wiley, 2017.
- [5] ISO 16610-21:2011, *Geometrical product specifications (GPS) — Filtration — Part 21: Linear profile filters: Gaussian filters*; superseded by ISO 16610-21:2025.
- [6] ISO 4288:1996, *Geometrical product specifications (GPS) — Surface texture: Profile method — Rules and procedures for the assessment of surface texture*; superseded by ISO 21920-3:2021.

A Statistical details

Model. With $y_{rjs} = \ln P_{\max}$ for rotor r (4), run j (2) and set point s (14),

$$y_{rjs} = \mu + \alpha_r + \beta_s + (\alpha\beta)_{rs} + u_{rj} + e_{rjs},$$

where u_{rj} is a run offset within the rotor and e_{rjs} the residual [4]. The rotor effect is tested against run within rotor (the one-way ANOVA on run means), and the interaction against the residual (Table 6). The interaction test assumes equal residual variance across set points, so it is approximate; it was repeated without the two set points with the largest scatter (500 and 900 rpm), $F_{33,44} = 23.7$, $p < 0.001$.

Table 6. Analysis of variance of $\ln P_{\max}$. MS in units of 10^{-4} .

Source	df	MS	F	p	Tested against
Rotor	3	1 550	324	$p < 0.001$	run within rotor
Set point (wind speed)	13	187 000			
Rotor $\times$ set point	39	31.7	9.22	$p < 0.001$	residual
Run within rotor	4	4.78	1.39	$p = 0.25$	residual
Residual	52	3.44			

Comparisons. With pooled run-to-run standard deviation $\sigma = 0.58\%$ (4 degrees of freedom), Tukey’s interval for rotors a and b is $\exp(\bar{y}_b - \bar{y}_a \pm q_{0.05;4,4} \sigma / \sqrt{2}) - 1$ with $q_{0.05;4,4} = 5.76$, a half-width of about 2.4%. An exact permutation test of the rotor effect over all 105 splits of the run means gives $p = 1/105$. The parabolic-fit estimator changes the differences by at most 0.5 percentage points.

Mounting variation. If each rotor’s mean carried an independent mounting offset with standard deviation τ , a difference would stay resolved while $|\bar{y}_b - \bar{y}_a| > (q_{0.05;4,4} / \sqrt{2}) \sqrt{\sigma^2 + 2\tau^2}$.

Drift. The run means are fitted as $\bar{y}_{rj} = a_r + \beta t_{rj}$, with t_{rj} the run start time in hours. The slope rests mostly (94%) on the two Plain runs, the only pair an hour apart.

B Surface analysis

Each image column of the Keyence height map is one profile across the layer lines. Columns with too many missing points are dropped (1.8% or fewer for Plain, FS 0.05 and FS 0.10; 5.1% for FS 0.20, whose values may therefore be understated). Each profile is levelled; P_a is its mean absolute deviation. R_a uses the Gaussian mean line of ISO 16610-21 [5] with $\lambda_c = 0.25$ mm over a 1.13–1.17 mm evaluation length. The layer period is the peak of the mean spectrum between 60 and 320 μm . ISO 4288 [6] recommends this cut-off for the 0.1 mm layer lines of FS 0.05 and FS 0.10, but longer ones for Plain’s 0.2 mm lines and for FS 0.20; both exceed the 1.4 mm field. R_a therefore keeps about 65% of Plain’s layer-line amplitude and 99% of the 0.1 mm ones.

C Wind-speed calibration

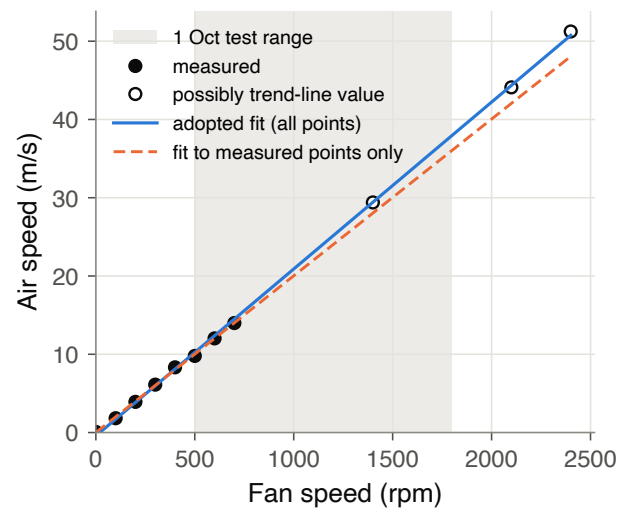


Figure 8. Tunnel calibration, Test 1 (13 February 2026). Filled: measured. Hollow: marked in the calibration file as possibly read from the trend line of the original record. Shaded: the range of this test.

Table 7. Calibration points and the adopted fit $v = 0.02132N - 0.424$.

Fan (rpm)	v (m/s)	Source	Fit (m/s)	Fit – meas. (m/s)
0	0.08	measured	−0.42	−0.50
100	1.83	measured	1.71	−0.12
200	3.92	measured	3.84	−0.08
300	6.12	measured	5.97	−0.15
400	8.33	measured	8.10	−0.23
500	9.79	measured	10.24	+0.45
600	12.03	measured	12.37	+0.34
700	13.99	measured	14.50	+0.51
1400	29.40	trend line?	29.43	+0.03
2100	44.10	trend line?	44.35	+0.25
2400	51.24	trend line?	50.75	−0.49

D Data package

The accompanying archive holds the raw files of the eight runs (`1_rig_sweeps/`), T. Kang’s rotor-speed summaries and RPM.m (`2_rotor_speed/`), the profilometer height maps (`3_surface_scans/`), the analysis-ready tables, including every run’s peak power and rotor speed at every set point (`4_derived/`), reference files (`5_reference/`), the data figures (`6_figures/`) and the analysis code (`7_code/`). `README.md` explains the layout and `DATA_DICTIONARY.md` every column. `7_code/build_report.py` regenerates every number in this report from the raw files.